ARJUN SHUKLA

Pune, India | arjunshukla6558@gmail.com | github.com/arjun988

EDUCATION

Pimpri Chinchwad College of Engineering Pune, India
Bachelor of Engineering in Computer Engineering; CGPA: 8.9/10.0 *2025*

Relevant Coursework: Data Structures, Software Engineering, Computer Networks, Operating Systems, Algorithms, Artificial Intelligence

TECHNICAL SKILLS

AI Frameworks & Tools: LangChain, LangFlow, LangGraph, Agno, LangFuse, LangSmith, Atlas Vector DB, Ollama, LlamaIndex, TensorFlow, OpenCV, NLTK, Pandas, NumPy
Frontend Technologies: React, NextJS, React Native, TypeScript, JavaScript, HTML, CSS, Tailwind CSS, AI Elements (UI)
Backend Technologies: FastAPI, Django, NodeJS, ExpressJS, Flask, Python, Fastify, Gunicorn, Nginx
Databases: MongoDB, PostgreSQL, Prisma, MySQL, Firebase, Oracle, NoSQL
AI Development: Deep Learning, Machine Learning, LLM Engineering, Natural Language Processing, Computer Vision, Data Science, Multimodal AI Pipeline
Cloud & Infrastructure: AWS, Azure, GCP, Docker, Kubernetes
Observability & Monitoring: LangFuse, LangSmith, HyperDX
Version Control: Git, GitHub, GitLab, Bitbucket
Developer Tools: Cursor, Claude Code, Kiro, NPM,

WORK EXPERIENCE

CrelioHealth Pune, India
Junior Software Engineer *October 2025 – Ongoing*

- Architected end-to-end AI-powered healthcare processing and lab inventory management system using React, FastAPI, Django, LangChain, Agno, Prisma, and Vector DB (Atlas), serving 2000+ laboratories and increasing operational efficiency by 200x compared to traditional software
- Led full-stack development with complete product ownership, implementing multimodal AI pipeline with LLM engineering, LangFlow, OAuth authentication, accelerating critical lab workflows and improving processing accuracy across frontend, backend, and database layers

CrelioHealth Pune, India
Engineering Intern *April 2025 – October 2025*

- Contributed to full-stack development of core healthcare platform using React.js, TypeScript, Python, and FastAPI, delivering scalable features and enhancements in a production environment
- Integrated advanced AI capabilities using LangChain, LangGraph, and AGNO to enhance product intelligence, while managing containerized microservices with Docker for efficient deployment and development workflows

Praudyogic AI Dubai, UAE
AI Engineer Intern *July 2024 – October 2024*

- Introduced an efficient LLM chatbot powered by OpenAI's API to assist clients in scheduling therapy appointments; optimized the process, resulting in a 25% increase in completed bookings
- Launched an integrated chatbot solution with the Zoho Bookings API that refactored the appointment booking process, now utilized by 12 customer service representatives, improving response time across the board

Celebal Technologies Pune, India
Data Science Intern *May 2024 – August 2024*

- Gained hands-on experience in analyzing large datasets of 100000+ data points and extracting meaningful insights using advanced data science techniques and tools
- Developed and implemented machine learning models to solve real-world business problems, enhancing predictive analytics and algorithm development skills

PROJECTS

AI Labs | ai-labs

- Developed no-code AI workspace with drag-and-drop ML workflow design, featured on Product Hunt, serving 400+ users and reducing AI experimentation time by 66%
- **Successfully sold the product to Frontier Labs**, demonstrating commercial viability and market demand for accessible AI development tools

AImake | GitHub

- Developed an open-source incremental build system for AI/ML and multi-agent pipelines, modeling datasets, prompts, embeddings, models, evaluations, and reports as a dependency graph with content-based invalidation
- Engineered content-addressable caching, parallel and GPU-aware execution, distributed workers, experiment optimization, and integrations across Hugging Face, DVC, Docker, and Ollama

Universal AI Inference Runtime (UAII) | GitHub

- Engineered a modular C++17 inference runtime that converts Hugging Face, GGUF, Safetensors, ONNX, MLX, and PyTorch models into a common intermediate representation (IR) for hardware-independent execution
- Implemented graph validation/planning, operator fusion, memory reuse, KV-cache generation, quantized MatMul, and weight streaming across CPU, CUDA, Metal, Vulkan, WebGPU, and ROCm backends
- Built plugin-based inference runtime with stable C ABI and Python bindings; reduced weight memory by 7.1x using Q4-0 quantization and achieved 2.77 ms median latency on an 8x512 MatMul+ReLU benchmark

StoryCraft | GitHub

- Built cross-platform mobile storytelling app achieving 30% UX improvement from 50+ beta testers, 25% engagement increase, and 35% DAU growth among 150+ users
- Deployed Django backend on AWS reducing data retrieval by 20% for 200+ concurrent users; architected Stable Diffusion image server; **sold AI infrastructure to Frontier Labs**

Euler - CLI Coding Agent | GitHub

- Developed personalized local CLI coding agent for terminal-based code generation, editing, and refactoring using natural language with multi-file operations support
- Implemented local LLM integration with file system operations, git integration, and intelligent code analysis for workflow automation without external API dependencies

RAT (Reinforced Adaptive Transformer) | GitHub

- Architected next-generation transformer with RL-powered adaptive attention, RoPE, SwiGLU networks, and temporal convolutions, installed by 5000+ developers
- Implemented multi-policy attention with dynamic head gating and comprehensive training pipeline featuring auto-checkpointing, KV caching, and optimized text generation

DCEE (Delta-Compressed Embedding Engine)| GitHub

- Developed Python library for ANN search over correlated embeddings achieving 60% memory reduction, 40% latency improvement, 95%+ recall accuracy, installed by 3000+ users
- Designed scalable vector search infrastructure outperforming traditional ANN libraries by 3x in memory efficiency for document chunking, log analysis, and conversational AI

ACHIEVEMENTS

Entrepreneurial Success

– Successfully sold AI Labs (first venture) and AI image generation server infrastructure to Frontier Labs, demonstrating commercial viability and technical innovation in AI product development

BOROSA: Bosch Road Safety Hackathon

– Earned recognition by securing 2nd runner-up position out of 170 teams and 450 students during the Bosch Road Safety Hackathon

Buildspace: Nights and Weekends

– Got selected in Buildspace Nights and Weekends Season 5 out of 32000 participants and teams across the globe to work on AI-based project 'StoryCraft'